

Pardhasaradhi Reddy Polaka

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OBJECTIVE

Experienced Data Engineer with over 6 years of expertise in Big Data and cloud platforms (AWS, Azure, GCP). Skilled in optimizing ETL pipelines, automating infrastructure, and integrating data across industries such as healthcare, finance, gaming, and e-commerce. Proven ability to enhance data processing efficiency by up to 20%, improving accessibility and analytical insights. Seeking to leverage cloud and big data engineering expertise to drive data-driven decision-making and innovation within a dynamic organization.

WORK EXPERIENCE

DATA ENGINEER | INTUIT, PLANO, TX

01/2023 – PRESENT

- ◆ Automated workflows for daily incremental loads using **AWS Step functions**, facilitating the smooth transfer of data from traditional **RDBMS** to **AWS Data Lakes**, such as **Amazon S3**.
- ◆ Enhanced **AWS Lambda functions** for optimized data extraction and transformation, reducing processing time by 20%.
- ◆ Engineered scalable data integration pipelines handling diverse sources like **electronic health records (EHRs)** and **claims data**, improving data ingestion speed by 10%.
- ◆ Developed and maintained data storage solutions using **AWS**, including **RDS** and **Redshift**, resulting in efficient data management.
- ◆ Executed ETL/ELT pipelines with **AWS Glue** and **AWS Step Functions**, increasing data processing efficiency by 20%.
- ◆ Automated workflows using **AWS Step Functions** for seamless daily data loads from **RDBMS** to **AWS Data Lakes**, achieving a 30% increase in data availability.
- ◆ Developed **shell scripts** for configuring and deploying applications across multiple **UNIX-based servers**, streamlining environment setup and reducing deployment time by 50%.
- ◆ Designed and implemented **GCP** services (Dataflow, Dataproc, Pub/Sub) for real-time and **batch data** transformation, reducing manual involvement by 85% with **Terraform**.
- ◆ Developed and optimized data pipelines using **PySpark** to process large volumes of structured and semi-structured data, improving data throughput and reducing latency by up to 30%.
- ◆ Employed **AWS Glue** and **Spark** for data cleansing and applied transformations, ensuring data analysis efficiency by 20% and reducing errors by 20%.
- ◆ Designed and implemented **RESTful APIs** using **Spring Boot** to enable seamless communication between micro-services, improving system scalability and reducing latency by 30%.
- ◆ Developed **GraphQL APIs** for dynamic querying, allowing clients to fetch specific data and reducing over-fetching issues, resulting in 25% faster response times.
- ◆ Executed workflow automation and coordination using **Apache Oozie**, complemented by dynamic and extensible workflow orchestration through **Apache Airflow**.
- ◆ Utilized **TensorFlow Extended (TFX)** for end-to-end pipeline orchestration, streamlining the preprocessing, training, and deployment of ML models.
- ◆ **Collaborated** with cross-functional teams for the seamless integration of the AWS cloud platform, optimizing data storage using services like **Amazon S3** and other relevant tools.
- ◆ Developed **Python**, **PySpark**, and **Bash** scripts to transform and load data across on-premises and AWS cloud platforms.
- ◆ Developed **Scala** scripts to enhance data processing capabilities and improve efficiency in **ETL** pipelines alongside **AWS Glue** and **Spark**.
- ◆ Utilized **TDD** to write tests before developing **ETL pipelines**, reducing debugging time by 20% and ensuring functionality met requirements from inception.
- ◆ Implemented **TDD(Test-Driven Development)** practices to develop scalable and reliable data engineering pipelines, ensuring minimal defects and improved code quality.
- ◆ Optimized data pipelines to ensure real-time data availability for **visualization tools**, improving the performance of dashboards and reducing query load time by 40%.
- ◆ Enhanced data analytics capabilities by implementing **Databricks** on **AWS**, contributing to increased accuracy and speed in data analysis, resulting in a 20% increase in data analysis speed and accuracy.

DATA ENGINEER | LOWES, HYDERABAD,INDIA

03/2019 – 08/2021

- ◆ Designed and implemented scalable **data pipelines using GCP Dataflow**, reducing data processing times for batch and streaming workloads by 40%.
- ◆ Optimized complex **BigQuery queries** for analytics workflows, achieving a 30% improvement in query execution time while minimizing cost.
- ◆ Streamlined data ingestion by 30% using **Apache NiFi** for ETL pipeline management from multiple sources (e.g., MS-SQL Server, PostgreSQL).
- ◆ Built **Snowflake data warehouse** for centralized storage and optimized structured and semi-structured data integration, boosting processing speed by 20%.
- ◆ Tuned **PySpark** jobs to optimize cluster resource usage, resulting in a 15% decrease in computation costs and improved processing speed on cloud platforms like **AWS EMR** and **Databricks**.
- ◆ Developed and optimized complex SQL queries in **Teradata** to process large-scale datasets, enhancing query performance by 25% and improving data retrieval times for critical business reports.
- ◆ Utilized **Teradata** utilities such as **FastLoad** and **MultiLoad** for efficient data ingestion and ETL operations.
- ◆ Reduced latency by 30% in data processing pipelines using **Golang** for real-time analytics on large datasets.
- ◆ Implemented real-time streaming with **Apache Kafka** and expanded **NoSQL** storage with **HBase** and **Cassandra**, enhancing throughput for large datasets.
- ◆ Implemented Big Data processing using **Hadoop** ecosystem components and enhanced real-time analytics capabilities by integrating **Apache Flink** for stream processing.
- ◆ Designed and implemented **end-to-end system integration workflows**, enabling seamless data flow between disparate enterprise systems and reducing manual intervention by 30%.
- ◆ **Jira/Confluence** was used to organize sprints and scrums, allowing the team to operate in an agile setting that increased **productivity** and **teamwork**.
- ◆ Leveraged **AWS Yarn** for resource management and integrated **Amazon EKS (Elastic Kubernetes Service)** for container orchestration, ensuring scalable and efficient data processing across the platform.
- ◆ Implemented robust data governance policies, access controls, and encryption mechanisms across the unified platform, utilizing **Amazon IAM** for identity management and **AWS key management service (KMS)** for encryption.
- ◆ Leveraged **Scala** for developing custom data processing algorithms and pipelines, enhancing flexibility and efficiency in handling diverse data sources and formats.
- ◆ Implemented feature engineering techniques using **Pandas** and **NumPy**, improving the performance of predictive models by 15%.
- ◆ Leveraged **NLTK** for natural language processing (NLP) tasks, including text tokenization, sentiment analysis, and entity recognition, to derive insights from unstructured data.
- ◆ Optimized **CI/CD** pipelines with automated testing frameworks to integrate **TDD practices**, reducing deployment errors by 30%.
- ◆ Established monitoring solutions using tools like **Grafana** and **Amazon CloudWatch**, seamlessly integrating performance tuning mechanisms, achieving a 10% improvement in system reliability.
- ◆ **Excel datasets** from many sources were combined and cleaned to increase **data accuracy** and facilitate more trustworthy analysis.

DATA ENGINEER | eBay, BANGALORE, INDIA

05/2017 – 03/2019

- ◆ Centralized **Data ingestion** from multiple sources into a scalable **Big data ecosystem**, increasing ingestion efficiency by 30%.
- ◆ Utilized **SSAS** for **data modeling**, achieving a 20% increase in database administration efficiency.
- ◆ Employed data quality checks using **Talend** and **Informatica**, addressing inconsistencies and expanding capabilities with **Python** for versatile data processing.
- ◆ Established monitoring with **MSBI** and **Big Data tools**, optimizing data security by 20% and improving pipeline performance with advanced **Excel** reporting.
- ◆ Administered and maintained databases with tools like **MSBI**, ensuring performance, security, and availability, 20% enhancement in database administration efficiency.
- ◆ Implemented **ETL** processes through **SSIS** to transform raw data for analysis.
- ◆ Utilized **Python** for additional data manipulation, expanding capabilities and contributing to a more versatile data processing approach.

- ◆ Assisted in setting up and maintaining data infrastructure, involving **servers, clusters, and cloud services**, with big data technologies.
- ◆ Extensive background in integrating and loading data into **Python** processes from a variety of sources, including **databases, APIs, and flat files**. Proficiency in scalability and performance optimization of data pipelines.
- ◆ Monitored data pipelines and infrastructure for performance using **MSBI and Big Data** monitoring tools.
- ◆ Leveraged advanced **Excel** techniques, such as charts, graphs, pivot tables, to deliver comprehensive weekly and monthly reports.
- ◆ Collaborated with cross-functional teams, including **Data Scientists, Analysts, and Business Stakeholders**.

TECHNICAL SKILLS

- ◆ **Programming Languages and Databases:** SQL, MySQL, NoSQL, Scala, Golang(GO), UNIX Shell Scripting, Python, Java, C#
- ◆ **Database Tools:** SQL, MySQL, PostgreSQL, MongoDB
- ◆ **Cloud Platform:** AWS (EC2, S3, SQS, Redshift, Glue, Lambda, DynamoDB, EMR, Kinesis, FireHose, IAM roles etc.), Azure (ADF, ADLS, Databricks), GCP (Dataflow, BigQuery, Cloud Storage, Dataplex, Data Fusion, Dataproc, Cloud Composer, DataPrep, Kubernetes, Cloud Pub Sub,)
- ◆ **Big Data:** Hadoop (MapReduce, HDFS, Pig), Kafka, Apache Hive, Splunk, Flink, Teradata, EMR, HBase, Cassandra
- ◆ **Data Warehouse:** Snowflake, Azure Synapse Analytics, AWS Redshift, Oracle Data Warehouse
- ◆ **ETL Tools:** Apache Spark, Azure Data Factory, AWS Glue, AWS Lambda, Pentaho, Informatica PowerCenter, Talend, Apache NiFi
- ◆ **Data Visualization:** Power BI, Tableau, AWS QuickSight, Grafana, Pentaho, Advanced Excel functions
- ◆ **Data Processing:** Azure Databricks, Spark
- ◆ **Data Storage:** Azure Data Lake Storage, Azure Blob Storage, AWS S3, Hadoop
- ◆ **Monitoring and Logging:** Azure Monitor, AWS CloudWatch
- ◆ **Version Control:** Git, GitHub, GitLab, Jenkins
- ◆ **Project Management:** Jira, Confluence, Hive, Atlassian, Microsoft Office
- ◆ **File Formats:** JSON, XML, YAML, Avro, ORC, CSV, Parquet, Protobuf, Excel, Log files

EDUCATION

- ◆ **Master of Business Analytics** | The University of Texas at Dallas **Dallas, Texas**
- ◆ **Bachelor of Computer Science Engineering** | JNTUH University **Hyderabad, India**